

Seeing the Real Me: Experiencing Real-Body Pass-Through Embodiment and Interaction

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ABSTRACT

Embodiment is fundamental to immersive VR, yet traditional avatars often suffer from perceptual mismatches. Pass-Through Embodiment (PTE) addresses this by integrating a stereoscopic live video feed of the user's body into the virtual environment. By combining depth-based segmentation with video-see-through streams, PTE provides a high-fidelity, photon-captured representation. We present the first public PTE demonstration, allowing users to perceive themselves while interacting with diverse virtual scenes. Participants can evaluate how their body, different edge rendering, and environmental contexts affect visual coherence in real-time. This demonstrates how PTE facilitates a naturally anchored sense of embodiment, bridging physical and virtual worlds through natural interaction.

Index Terms: Virtual Reality, Embodiment, Video Pass-Through, Rendering

1 INTRODUCTION

A convincing sense of embodiment in Virtual Reality (VR) is critical for creating immersive experiences, shaping how users perceive and interact with their virtual selves. Traditional avatar-based approaches often rely on pre-defined or reconstructed models which often suffer from perceptual mismatches that degrade self-presence and agency [7, 3, 8]. Such discrepancies between the user's physical actions and virtual representation can disrupt the sense of being truly located within the digital environment. Pass-Through Embodiment (PTE) provides a compelling alternative by integrating a live video feed of the user's physical body directly into the virtual scene [6]. Utilizing modern Head-Mounted Displays (HMDs) with depth-sensing capabilities, such as the Meta Quest 3, PTE enables immediate and realistic self-representation. Research has shown that providing an egocentric view of one's own physical body via pass-through significantly improves perceptual congruence, leading to improved presence and embodiment compared to traditional avatars [6, 2, 1].

In this work, we build upon our previous research [6, 5] and can present an interactive demonstration of PTE that allows participants to experience their real body in three distinct immersive virtual environments (IVEs) while evaluating various cutout edge-rendering techniques.

Our contribution is an interactive platform designed to explore how specialized boundary rendering and real-time hand-to-object

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Figure 1: Egocentric views of the Pass-Through Embodiment (PTE) method in distinct virtual environments (see more in Fig. 2). The system utilizes HMD depth-sensing and video pass-through. The *left* image illustrates a downward view of the user's physical body, while the *right* image demonstrates real-hand interaction with a virtual object.

interaction facilitate a naturally anchored and perceptually stable sense of embodiment. By demonstrating the PTE method across varied virtual contexts, we enable users to directly experience the impact of different rendering techniques on visual coherence and self-recognition.

2 METHOD

2.1 Concept of Pass-Through Embodiment

Pass-Through Embodiment (PTE) integrates the user's physical body into a virtual environment by leveraging the video pass-through and depth-sensing capabilities of modern Head-Mounted Displays (HMDs). Building upon our previous work [5, 6], this method bypasses the need for complex avatar modeling or motion capture. Instead, users perceive their actual physical selves, which significantly enhances self-recognition and spatial presence. Our current implementation focuses on refining the boundary transitions and spatial anchoring required to create a seamless immersive experience.

2.2 System Features and Rendering

The demonstration incorporates several core features designed to optimize the PTE experience. We adopt the three rendering modes evaluated in our previous work [5] to treat the cutout boundary between the physical body and the virtual scene: **Hard Cutout** for sharp but potentially aliased edges, **Soft Cutout** for smooth gradient blending, and **Dithered Cutout**. The latter utilizes high-frequency noise patterns to perceptually bridge the gap between

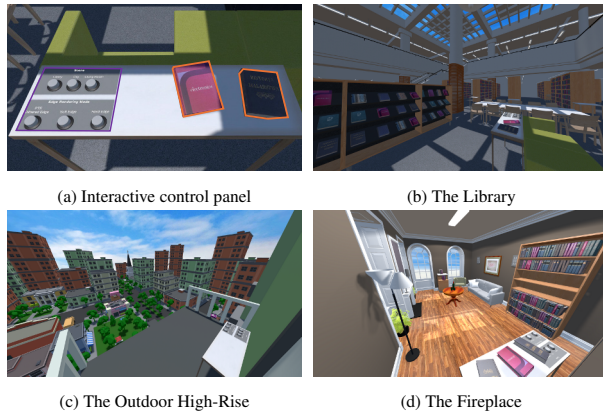


Figure 2: (a) A table with a control interface enables users to toggle between various rendering modes and environments while also providing a place for interactive objects, such as books. (b–d) Virtual scenes used in our demonstration to experience PTE.

edge clarity and smooth transitions. Further technical details regarding the implementation and perceptual evaluation of these features can be found in the full paper [5]. To enhance spatial anchoring, we utilize shadow proxy meshes [6] that receive virtual shadows and provide occlusion cues. In this demonstration, we specifically focus on hand proxies to ensure the primary interfaces for interaction are precisely grounded, facilitating accurate manipulation and multi-hand support.

2.3 Demonstration Overview and Scene Design

We chose to have three distinct virtual environments, each selected to investigate the interaction between scene context and PTE (see Fig. 2). Our system operates via controller-free hand tracking, enabling natural gestures and fluid movement within each setting. To create a seamless experience, a virtual control table is positioned within each environment (see Fig. 2a). This interface features two sets of interactive buttons: three for toggling between the different rendering modes and three for immediate scene transitions. This allows participants to switch and compare the perceptual impact of environmental complexity and edge rendering techniques in real-time. We chose to have the following three environments:

The Library: An indoor environment with high-frequency geometric detail and artificial lighting. It encourages participants to inspect their body representation closely against familiar, dense surroundings.

The Outdoor High-Rise: A low-poly, stylized balcony scene. The verticality and openness of this environment highlight spatial perception and test how cartoon-inspired aesthetics modulate the salience of pass-through artifacts.

The Fireplace: A reference scene from the *McGuire Computer Graphics archive* [4]. This serves as a standardized benchmark to evaluate rendering quality and embodiment in a controlled, familiar context used in prior research.

3 DISCUSSION

As a result, PTE provides a viable alternative to avatar-based methods by eliminating the need for complex animation, while enhancing presence and self-recognition [6]. Our results across diverse environments show that no single edge treatment is universally optimal. While rendering modes exhibit perceptual differences [5], preferences remain varied. We hypothesize that scene context and visual consistency are decisive, as high-frequency geometry may accentuate boundary artifacts such as the *Library* scene.

Despite these advancements, certain limitations remain. The lighting mismatch between the video pass-through of the physical body and the synthetic illumination of the virtual environment remains a primary challenge. Additionally, the reduced resolution of the depth map can cause spatial inaccuracies, particularly leading to z-fighting artifacts on aligned flat surfaces like floors. To mitigate this, the virtual floor is often slightly elevated above the physical geometry, which can result in the partial clipping of the user's feet. Although edge dithering introduces negligible computational overhead [5], the inherent video pass-through delay of approximately 40 ms on the Meta Quest 3 must be considered, as it impacts the overall temporal consistency of the embodiment.

4 CONCLUSION

We present the first PTE demonstration integrating diverse virtual environments with interactive edge rendering. By enabling users to perceive their physical bodies across contexts, we show how scene design and boundary treatments modulate visual coherence. The system further highlights PTE's potential for natural interaction and self-recognition, establishing a naturally anchored alternative to traditional avatars for immersive VR.

For future work, we will focus on performance optimizations, as the current requirement for depth checks across all scene objects introduces shader-side bottlenecks, as well as evaluating PTE further and how its visual congruency affects overall coherence in different scenarios.

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